INTELIGENTNA SZAFKA – STM32 + ESP32 / 2 szuflady

Power_12V_5V_3V3

Wejscie 12V DC / SS54 / TVS
AP63205: 12V -> 5V_BUCK
SS14 OR: USB i buck -> 5V_OR
AP63203: 5V_OR -> +3V3

File: Power_12V_5V_3V3.kicad_sch

STM32_Control_SWD

STM32G030C8T6 / LQFP48
SWD STDC14 / UART do ESP32
PWM napedow / ADC IPROPI
I2C ToF / krańcówki / status

File: STM32_Control_SWD.kicad_sch

ESP32_USB_Audio

ESP32-S3-WROOM-1-N8
Natywne USB-C / ESD / 22R
Przyciski BOOT i RESET
I2S do zewnetrznego INMP441

File: ESP32_USB_Audio.kicad_sch

Motor_Drivers

2x DRV8876RGTR (PINOUT TO VERIFY)
PWM / IPROPI / VREF wstepne
Bramka AND steruje nSLEEP
Dwa niezalezne wyjscia silnikowe

File: Motor_Drivers.kicad_sch

Safety_Sensors_Connectors

NC: E-STOP + kontakty obu szuflad
AND: MCU_ENABLE && SAFETY_OK
4 osobne krańcówki NC
2x ToF: I2C + oddzielne XSHUT

File: Safety_Sensors_Connectors.kicad_sch

Arkusze lacza etykiety globalne. Zasilanie: 12V -> VM_MOTORS; USB tylko 5V_OR / +3V3.

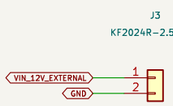
SCHEMAT ROBOCZY. Brak weryfikacji katalogowej nie jest wynikiem ERC. Bez PCB/Gerber/CPL.

NOT FOR ORDER – manufacturer/catalog/actuator verification pending

Sheet: / File: Szafka_Glosowa_STM32_ESP32.kicad_sch		
Title: Szafka_Glosowa_STM32_ESP32		
Size: A3	Date:	**Rev: DRAFT 0.1**
KiCad E.D.A. 9.0.2		Id: 1/6

Power_12V_5V_3V3

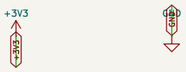
DRAFT – dane katalogowe i footprinty do potwierdzenia; NIE DO ZAMOWIENIA





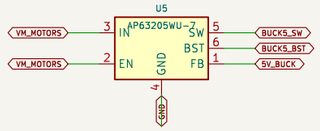






Zewnetrzny bezpiecznik DC TBD przed J3. Na PCB tylko 12V DC.

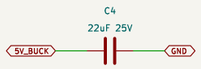
D1 straty/termika i D4 koordynacja z bezpiecznikiem: TO VERIFY.

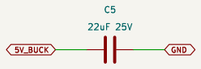


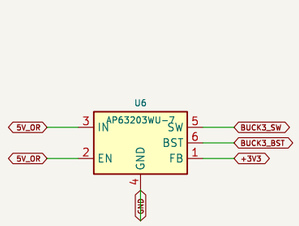








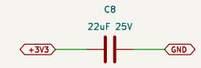


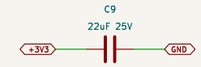


















USB -> D3 -> 5V_OR -> U6 -> +3V3. USB nie ma drogi zasilania VM_MOTORS.

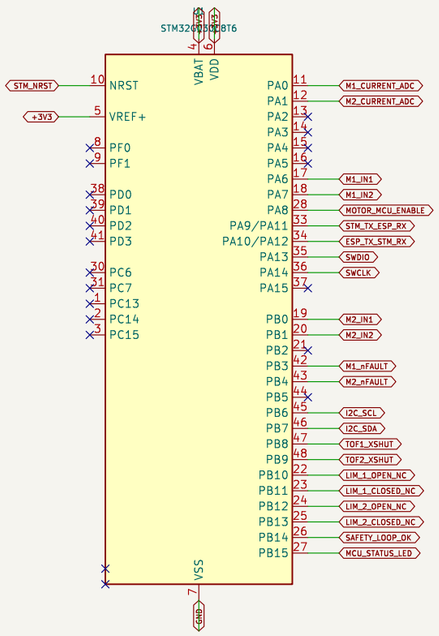
U5/U6: stale napiecia; FB do wyjscia ZA dlawikiem. CIN/COUT i BST blisko ukladow.

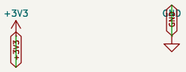
WORKBOOK INPUT; datasheet/catalog verification pending; NOT FOR ORDER

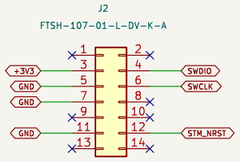
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Size: A3	Date:	**Rev: DRAFT 0.1**
KiCad E.D.A. 9.0.2		Id: 2/6

STM32_Control_SWD

DRAFT – dane katalogowe i footprinty do potwierdzenia; NIE DO ZAMOWIENIA





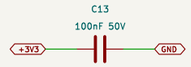


STDC14 wg arkusza: 3 VTref, 4 SWDIO, 6 SWCLK, 12 NRST.

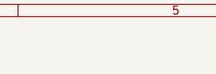
1/2/9 reserved; 8/10/13/14 nieuzywane. J2 footprint do weryfikacji.

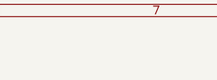






C11 VDD, C12 VBAT, C13 VREF+; blisko odpowiednich pinow U1.





UART: PA9/PA11 = pin33, PA10/PA12 = pin34; piny29/32 NC.

Sprawdz remap PA9/PA10 i AF w datasheet/reference manual przed firmware.

HSI bez kwarcu. Nieuzywane GPIO oznaczone NC; krańcówki wymagaja pull-up w firmware.









WORKBOOK INPUT; datasheet/catalog verification pending; NOT FOR ORDER

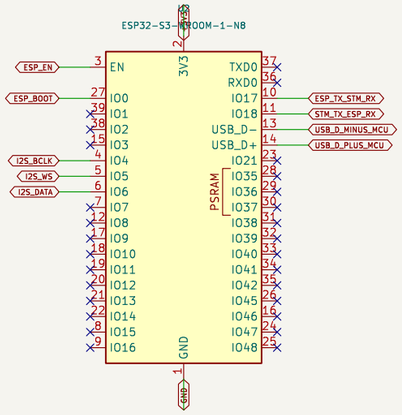
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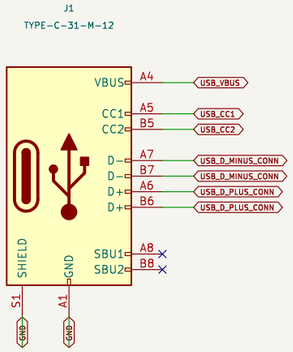
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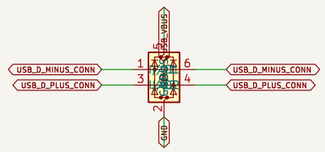
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KiCad E.D.A. 9.0.2		Id: 3/6

ESP32_USB_Audio

DRAFT – dane katalogowe i footprinty do potwierdzenia; NIE DO ZAMOWIENIA





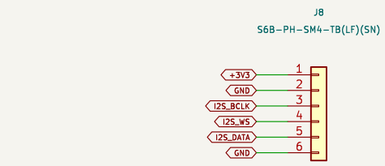


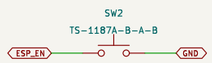


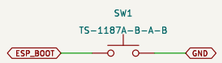








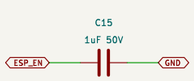












INMP441 zewnetrzny: pin6 L/R = GND, lewy kanal. Potwierdz kabel/modul.

USB GPIO19/20 = piny modulu13/14; UART GPIO17/18 = 10/11.

J1: sygnaly SMT, kotwy oslony THT! PCBA i wymaganie SMD do uzgodnienia.

22R wstepne; antena RF keepout, impedancja USB i ESD przy J1 do layout review.

WORKBOOK INPUT; datasheet/catalog verification pending; NOT FOR ORDER

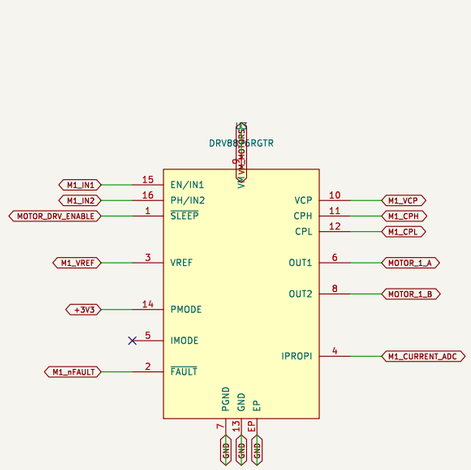
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File: ESP32_USB_Audio.kicad_sch

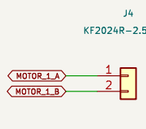
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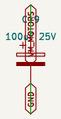
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KiCad E.D.A. 9.0.2		Id: 4/6

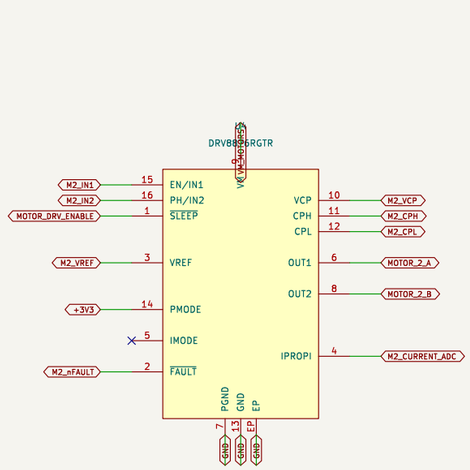
Motor_Drivers

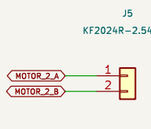
DRAFT – dane katalogowe i footprinty do potwierdzenia; NIE DO ZAMOWIENIA

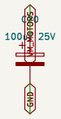


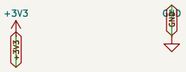












RGT z mapy XLSX; pin EP i land pattern NIEPOTWIERDZONE.

IMODE Hi-Z zamierzone wg XLSX; PMODE high (PWM).









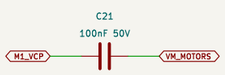




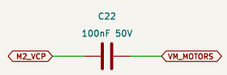




















VREF = 3.3V * 10k/(6.8k+10k) = 1.964V. WARTOSC TYLKO WSTEPNA.

RIPROPI/VREF i wspolczynnik sense musza byc sprawdzone dla wybranego silnika.

WORKBOOK INPUT; datasheet/catalog verification pending; NOT FOR ORDER

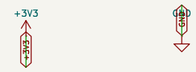
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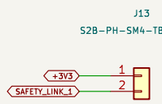
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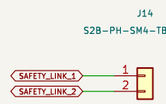
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KiCad E.D.A. 9.0.2		Id: 5/6

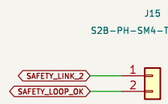
Safety_Sensors_Connectors

DRAFT – dane katalogowe i footprinty do potwierdzenia; NIE DO ZAMOWIENIA



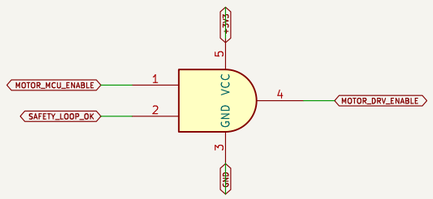






J13 E-STOP NC -> J14 kontakt szuflady1 NC -> J15 kontakt szuflady2 NC.

Rozwarcie styku obniza SAFETY_LOOP_OK przez R8, niezaleznie od kodu STM.











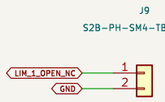


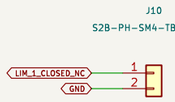


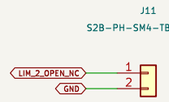


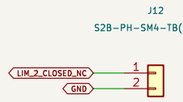
R7 + R8 default OFF; R18 dodatkowy pulldown nSLEEP; C28 bypass U8.

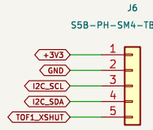
Prototyp NC/ToF/IPROPI NIE jest certyfikowanym ukladem ochrony BHP.

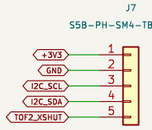












WORKBOOK INPUT; datasheet/catalog verification pending; NOT FOR ORDER

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File: Safety_Sensors_Connectors.kicad_sch

Title: Szafka_Glosowa_STM32_ESP32 / Safety_Sensors_Connectors

Size: A3	Date:	Rev: DRAFT 0.1
KiCad E.D.A. 9.0.2		Id: 6/6